

Tharidu Deshan Ekanayaka

Software Engineer | Full-Stack Development and AI Systems

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SUMMARY

Software engineer (BSc Computer Science, First Class Honours, University of Westminster, Jun 2026) building full-stack applications and production AI systems, from RESTful APIs to LLM-backed agent workflows.

Rebuilt a clinical decision support system to 94% Random Forest accuracy across 662 patients, resolving 6 examiner-identified failures in authentication, architecture, and ML validity.

Currently developing server-side AI systems, RAG pipelines, and agent workflows at FlyRank; seeking graduate software engineering and junior AI/ML roles in London.

SKILLS

Core Competencies: Full-Stack Development | JavaScript | TypeScript | Python | REST API Design | Agile/Scrum | Automated Testing | LLM Integration

Languages: JavaScript | TypeScript | Python | SQL | Java | HTML5 | CSS3

Frameworks: React | Next.js | Node.js | Express.js | FastAPI | Django | Spring Boot | TailwindCSS

Data and ML: PostgreSQL | MySQL | SQLite | scikit-learn | pandas | NumPy | Anthropic and OpenAI APIs | RAG and Agent Workflows

Infrastructure and Tooling: Git | Docker | GitHub Actions CI/CD | Zod | Pydantic | Swagger/OpenAPI

AI-Assisted Development: Claude and CodeRabbit integrated into daily engineering workflow for debugging and PR review; all architecture, security, and clinical-threshold decisions authored manually.

PROJECTS

Diacify – Clinical Decision Support System

Sep 2025 – Apr 2026

React 18, Node.js/Express, FastAPI, MySQL, scikit-learn, Docker, GitHub Actions

diacify.vercel.app

- Built a web application that helps clinicians prioritise diabetes patients by risk, combining patient management, appointments, and a machine learning model that flags high-risk cases.
- Rebuilt the system from scratch, resolving 6 examiner-identified security and architecture failures, and architected a decoupled three-service system (React/Vite frontend, Node/Express API with Zod validation, FastAPI classifier microservice).
- Trained a Random Forest on 662 patients across 13 clinical indicators, improving accuracy from 89% to 94% via GridSearchCV; enforced cryptographic session-token authentication on every backend route.
- Implemented 3 GitHub Actions CI/CD workflows and wrote 8 Jest and 4 pytest tests, normalising a flat schema into patients, visits, appointments, and audit_log tables.

ClaimLens – AI-Powered Damage Claim Triage

Jun 2026 – Present

Python, FastAPI, Anthropic API (Claude Opus + Sonnet), Next.js 15, SQLite

github.com/deshanekanayaka/claimlens

- Built an AI tool that reads insurance damage claims and photos, checks evidence for contradictions, and recommends approve, reject, or escalate with a full audit trail; evolved from a HackerRank Orchestrate hackathon agent ranked top 13% (#234 of 1,773).
- Architected a three-call LLM pipeline (Claude Opus for extraction and synthesis, Claude Sonnet for image analysis) with deterministic evidence gating, and diagnosed 8 defects through a 12-scenario manual red-team plan.

EXPERIENCE

FlyRank

Remote

Backend AI Engineering Intern

Jul 2026 – Present

- Building the server side of AI products: systems that let large language models retrieve company data, call tools, and return reliable structured output.
- Building RAG pipelines and tool-calling agent workflows in FastAPI against Anthropic and OpenAI models on Supabase/PostgreSQL, with structured-output enforcement via Zod and Pydantic and an evaluation harness for failure-mode analysis.

EDUCATION

University of Westminster

London, UK

BSc (Hons) Computer Science, First Class Honours

Sep 2023 – Jun 2026

- Core modules: Object-Oriented Programming, Algorithms, Client-Server Architectures, Cybersecurity, Artificial Intelligence.